

Touch3D- Exploring and reconstructing different object surfaces using tactile sensor.

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Abstract—Tactile exploration is essential for perceiving object structures in robotic tasks like grasping and manipulation. However, it remains challenging due to the limited sensing range of tactile sensors and the complexity of unknown environments. Three-dimensional (3D) reconstruction from tactile feedback enables robots to perceive complex object geometry in conditions or environments with limited visual capabilities; i.e., in conditions where vision is unreliable or occluded. This project introduces Touch-3D, an end-to-end pipeline that couples a simulated DIGIT tactile sensor with a reinforcement-learning (RL) exploration policy(agent) designed to efficiently reconstruct object surfaces within a limited number of interactions in a TACTO [2] simulation environment. The agent learns to explore and reconstruct unknown objects by intelligently selecting a sequence of touch actions. The methodology employs a Proximal Policy Optimization (PPO) algorithm to train the agent in a simulated PyBullet environment. By actively guiding the exploration process, our method incrementally gathers tactile data to reconstruct 3D object shapes, which can also serve as informative representations for downstream tasks. Despite being trained solely on basic geometric shapes, our approach generalizes to unseen YCB objects, achieving highest IoU coverage of 95%.

I. INTRODUCTION

Robots operating in unstructured environments frequently encounter situations where visual cues are important, and visual data can be degraded by poor lighting, occlusion, or blind spots easily. The ability to perceive and understand the three-dimensional structure of objects, where finer details such as curvature or small deformation are important, is fundamental for autonomous robotic systems. While visual sensors like cameras and LiDAR are frequently used, tactile sensing offers a complementary modality in poor lighting scenarios with occlusions, but especially when direct physical interaction is necessary. Tactile sensing is light-agnostic; it provides direct contact with the surface, which can help reveal finer details such as object structure or different materials. However, turning sequences of tactile readings into complete 3-D object models remains challenging because the sensor must be moved intelligently across the surface, managing the trade-off of exploration coverage against redundant or unsafe contacts or repeated contacts at the same place.

This project addresses the challenge of active 3D object reconstruction using tactile exploration. The goal of this project is to develop an RL agent that autonomously explores an object’s surface with a simulated tactile sensor, incrementally

reconstructing 3D point clouds in simulation. The objectives are:

- **Adaptive Exploration:** Implement an RL algorithm(PPO) to train an agent that learns a policy for an optimal exploration that decides where the next touch would be based on most recent tactile observations.
- **Reward function:** To define a reward structure that encourages both exploration of new surfaces and maximization of reconstruction quality (coverage).
- **Accurate Reconstruction:** Fuse the depth maps from most recent individual touches into a global model to quantify coverage.
- **Generalization:** To evaluate the agent’s performance in terms of surface coverage and the quality of the generated 3D point cloud on unseen shapes.

II. LITERATURE SURVEY

Active SLAM involves traversing an unknown environment while simultaneously localizing and constructing a map [4], [6]–[8], [12]. In this traditional SLAM approach, revisiting the marked areas is beneficial for correcting estimated localization errors [15], [20]. However, our goal is to design an efficient exploration pipeline that minimizes revisits and penalizes them. Active SLAM is generally formulated as a Partially Observable Markov Decision Process (POMDP) [20], with various reward function formulations. Unlike the conventional SLAM setting, which operates in 2D spaces, our project focuses on exploring a 3D workspace with limited sensing space.

Early efforts in tactile shape reconstruction relied on exhaustive scans with multi-finger arrays or mechanical probes, limiting practicality in time-critical tasks. With the introduction of soft, camera-based tactile sensors(e.g, Gelsight, DIGIT), researchers began exploiting depth cues. The field of robotic tactile sensing and object reconstruction has seen significant advancements in recent times. For instance, [17] trained on A3C (Asynchronous Advantage Actor Critic) agent in a three-dimensional maze, which incorporated LSTM (Long Short-Term Memory) to provide the memory capability. [10], [21]. Tactile sensors have been used in many different applications [9], [13], [19]. When robots come into contact with deformable objects, tactile sensors offer valuable information that enhances the robustness of the system alongside the vision sensors. Additionally, tactile information has been used

to estimate the pose of the objects [3], [5], [22] or the relative pose of the gripper. Similarly, tactile sensors have been employed to identify or reconstruct unknown objects [11], [14], [16], [18].

In the context of tactile exploration, Reinforcement Learning(RL) allows for an agent to learn an optimal policy by interacting with its environment and receiving feedback in the form of rewards. [ActXplore] RL agents can learn policies to actively select viewpoints or touch points to maximize information gain or reconstruction quality. Some of the algorithms include Deep Q-Networks (DQN), Proximal Policy Optimization (PPO), and Soft Actor-Critic (SAC). These algorithms have been applied in the related domains for tasks such as tactile localization, surface following and shape recognition.

Proximal Policy Optimization(PPO) was selected for this project as it offers stable policy updates and due to its ease of implementation, balancing sample efficiency, native support in Stable-baselines3 and a stable performance for continuous action spaces. PPO is an on-policy algorithm that optimizes the policy directly by iteratively collecting a batch of experiences and updating the policy based on an objective function that clips the probability ratio between the new and old policies. This clipping mechanism helps to prevent excessively large updates, which leads to more stable training and is ideal for continuous control.

III. METHODOLOGY

This project employs a reinforcement learning approach to train an agent for 3D object reconstruction using simulated tactile sensing using the TACTO [2] simulation environment. The core components of the methodology include the simulation environment, the simulated tactile sensor model, the agent’s architecture, the learning algorithm and the reward structure.

A. Pipeline Overview

The overall workflow can be summarized as follows:

- **Initialization:**

- 1) Loading the 3D object(a primitive shape) into the simulation environment(via PyBullet- a gym environment).
- 2) The DIGIT tactile sensor is initialized in simulation via TACTO simulation environment.
- 3) Initialize the initial point of contact on the object—it is predefined. This will act as first touch on the object.
- 4) The PPO-based RL agent is initialized with its policy and value networks.

- **RL Interaction:**

- 1) **Sense:** The agent receives its state, a concatenated stack of most recent tactile images.
- 2) **Act:** Agent’s policy network processes state and selects appropriate action.
- 3) **Reconstruction:** Based on the contact tactile point cloud is extracted and added to global reconstruction map.

- 4) **Reward:** Reward is calculated based on contact, novelty, penalties and coverage(The agent gets feedback based on how good the contact was, how novel it is, and how well it covers the object).
- 5) **Learning:** Agent updates the policy using PPO.

- **Termination:**

- 1) The episode ends if the exploration reaches 5000 evaluation steps(one episode) or the sensor moves too far off the object(goes out of bounds). This helps keep training stable and bounded.

Fig 1. provides an overview of the complete pipeline:

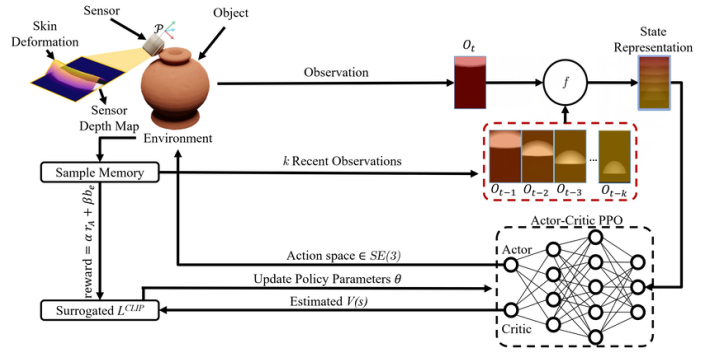


Fig. 1: **Overview:** This figure illustrates the key steps and components of Touch3D in a scenario where the sensor moves upward along the jar’s edge. We employed **Temporal Tactile Stacking** for state representation to encode consecutive observations, enabling the perception of movement on the sensor vital for learning dexterous actions. We also incorporate an Upper Confidence Bound (UCB) exploration as a bonus to encourage effective exploration.

B. Detailed Methodology

The detailed methodology is as follows:

- **Simulation Environment(PyBullet):** The project utilizes the PyBullet physics simulator to create an environment for interaction. Fig 2. showcases the 4 primitive shapes used in training:

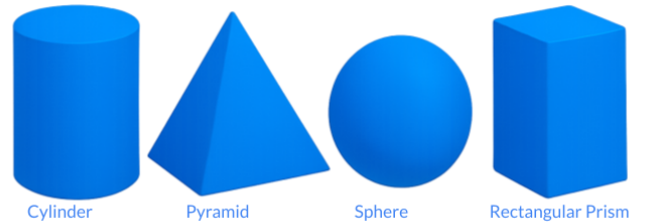


Fig. 2: Primitive Shapes

- 1) **Objects:** Four primitive 3D shapes: Sphere (radius 0.03m), cube (0.06m edge), pyramid (0.06m base, 0.08m height), cylinder (0.06m diameter, 0.08m height). These objects are loaded from URDF (Unified Robot Description Format) files. One object is randomly selected during each episode.
- 2) **Tactile Sensor:** A simulated version of the DIGIT tactile sensor is used. The tacto [2] library is employed to simulate the sensor's behavior, generating depth images(optionlly color images and point clouds) upon contact with the object. Sensor is mounted on a simulated fingertip like end-effector.
- 3) **Workspace:** Workspace boundaries are dynamically computed from the object's axis-aligned bounding box to constrain the movement of the tactile sensor and episode is terminated if the sensor moves out of bounds.

• **Agent and Action Space:**

- **Agent:** The RL agent's goal is to learn a policy $\pi(a_t|s_t)$ that maps the current state s_t to an action a_t to maximize expected future rewards.
- **Action Space:** The agent has a discrete action space consisting of 13 possible actions, which correspond to:

$$A = \{\pm x, \pm y, \pm z, \pm \gamma, \pm \theta, \pm \psi\},$$

where (x, y, z) are Cartesian translations with step size $T_s = 15mm$, and (γ, θ, ψ) are Euler angle rotations with $R_s = 15$. We augment this with a *Touch Recovery* action a_{TR} , which returns the sensor to the most recent valid contact pose. This yields a total of 13 discrete actions:

$$|A| = 12 + 1 = 13.$$

- **Observation Space (Space Representation):** The agent's observation (state) is derived from the tactile sensor's output. At each timestep t , the tactile sensor produces an image of skin deformation, which can be converted into a depth map $O_t \in \mathbb{R}^{H \times W}$. This depth image captures local surface geometry at the contact patch. We clip and normalize values to $[-1, 1]$ to stabilize learning.

To overcome the ambiguity of single-frame observations, the short-term history of the last k frames(depth images) is concatenated to form a state along the width dimension. This provides a short temporal history directly in the observation.

$$s_t = f(O_t, O_{t-1}, \dots, O_{t-(k-1)}).$$

This is how TTS is instantiated:

$$s_t = [O_{t-(k-1)} \parallel O_{t-(k-2)} \parallel \dots \parallel O_t] \in \mathbb{R}^{1 \times H \times kW}$$

• **Reinforcement Learning Algorithm(PPO):**

We employ Proximal Policy Optimization(PPO), an actor-critic method implemented via the stable-baselines3 library.

1) **Policy Network:**

CNN based actor-critic model s employed. The actor learns a probability distribution over actions, allowing exploration. The critic estimates the value of a given state which helps compute the advantage and improve sample efficiency.

As the agent's observations are image-based(TTS), CNN actor critic policy is well-suited for processing image data as it can effectively learn spatial hierarchies of features. Once the policy receives the tactile image observations of shape $\in \mathbb{R}^{1 \times H \times kW}$, Convolutional Feature Extractor extracts the relevant features from the input image.

By using a CNN, the agent can automatically learn to extract meaningful information from the raw pixel data of the tactile sensor, which is then used by the PPO algorithm to learn an effective exploration policy.

2) **Training Process:**

- The agent interacts with the TACTO environment by taking actions.
- For each action, it receives an observation, a reward, and a done signal.
- Experiences (state, action, reward, next state, done) are collected over a certain number of steps.
- The PPO algorithm then updates the policy and value networks using this batch of experiences to maximize a clipped surrogate objective function and minimize the value loss.

• **Reward Function:**

1) Reward Type: **AMB (Area Maximization with Bonus/Exploration):**

- The primary reward is based on contact area detected by the tactile sensor. A larger contact area yields a higher positive reward.
- A penalty is given if the agent revisits a state (or a pose very close to a previously visited pose that yielded a positive reward). This discourages redundant touches.
- **AMB** combines area reward with an exploration bonus. The exploration bonus can be written as: $R_{\text{explore}} = \frac{1}{\sqrt{N(s,a)}}$ where $N(s,a)$ refers to number of times the current (pose, action) pair has been visited. This encourages to try novel actions from novel poses.
- A penalty is applied if the sensor doesn't touch the object for a certain number of steps or repeated touches within 75% of translation/rotation step.
- The final reward is weighted sum of area reward and exploration reward. This is given by:

$$r = 0.15r_{\text{area}} + 0.85r_{\text{explore}}$$

- 2) A large positive reward is given if the coverage exceeds 90% to incentivize high completion.

We adopt the Area–Memory–Bonus (AMB) reward from AcTEExplore [1], encouraging wide surface coverage, novelty, and contact continuity. The reward is:

$$r(s_t, a_t) = \begin{cases} \alpha r_A + \frac{\beta}{\sqrt{\hat{N}(P_t, a_t)}}, & \text{if } r_A > 0 \text{ and } P_{t+1} \notin \mathcal{D}, \\ P_{\text{rev}}, & \text{if } r_A > 0 \text{ and } P_{t+1} \in \mathcal{D}, \\ P_{\text{TR}}, & \text{if } a_t = a_{\text{TR}}, \\ 0, & \text{otherwise} \end{cases}$$

where: itemsep=1pt

- r_A : contact area reward (fraction of nonzero depth pixels)
- $\hat{N}(P_t, a_t)$: count of similar pose-action pairs in history
- $\mathcal{D}$: short-term memory of recent poses (size $m = 20$)
- $P_{\text{rev}} = -0.03$: penalty for revisits
- $P_{\text{TR}} = -0.2$: penalty for touch recovery
- $(\alpha, \beta) = (0.15, 0.85)$: weight coefficients

This reward formulation balances **exploitation** (contact maximization) and **exploration** (novelty-seeking), while penalizing trivial or repetitive behaviors.

• 3D Reconstruction:

When the tactile sensor makes contact, the local point cloud from the sensor is transformed into the world frame using sensor pose. New points are only added to the aggregated point cloud if they are in the unoccupied vortex of the grid. This ensures only novel points contribute to the reconstruction. Coverage is measured as the fraction of ground-truth (GT) mesh vertices within 5mm of any reconstructed point, evaluated online via an Open3D KD-tree. Chamfer distance (bidirectional mean) provides a complementary accuracy metric.

• Training:

Learning Setup:

- Algorithm: **PPO** (learning rate: 0.0003, ent-coef: 0.00 num-steps: 2048 batch-size: 64 n-epochs: 10 gamma: 0.99 gae-lambda: 0.95)
- Policy Network: **CNN Policy network encoder**
- Reward: **TTS** (visited state penalty: -0.10, novelty threshold: 4, no touch threshold: 30)
- Training Steps: till 400k evaluation steps
- Logging: TensorBoard Metris
- Checkpoints: model saved every 50k steps.

– Finetuning:

- 1) **PPO** (learning rate: **0.0001**(reduced), ent-coef: **0.01**(encourages exploration) num-steps: 2048 batch-size: 64 n-epochs: 10 gamma: 0.99 gae-lambda: 0.95)
- 2) Policy Network: **CNN Policy network encoder**
- 3) Reward: **TTS** (visited state penalty: **-0.15**, novelty threshold: **3**, no touch threshold: **25**)
- 4) Training Steps: till **1.5M** evaluation steps
- 5) Logging: TensorBoard Metris
- 6) Checkpoints: model saved every 50k steps.

IV. CONTRIBUTION

• Repositories referred:

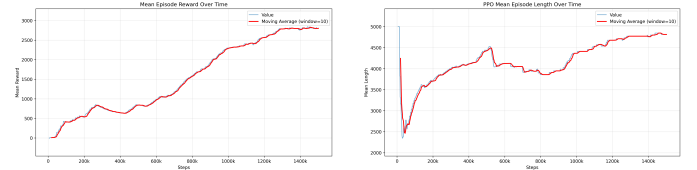
- For simulation enviroment and the TACTO sensor:
 - * **TACTO**: A Fast, Flexible and Open-source Simulator for High-Resolution Vision-based Tactile Sensors
 - * gihub link <https://github.com/facebookresearch/tacto>
- For base training algorithm and RL approach:
 - * **ActExplore**: Active Tactile Exploration on Unknown Objects
 - * gihub link <https://github.com/prgumd/ActTEExplore>

• Changes made to ActExplore repository:

- **Policy Network**: Changed from MlpPolicy to CN-NPolicy. The change is made to incorporate the tactile observations which are image-like sequences. CNNs excel at extracting spatial features(edges, shapes, textures) from images, which is crucial for interpreting contact information.
- **Input State Representation**: Change in the shape of concatenated depth maps to include channel dimension, $1 \times H \times (W \cdot 5)$.
- **Reward Function**: Refined the reward structure and termination logic; Improved reward calculation and logging for coverage milestones, Added more robust handling of out-of-bounds and step limits, inclusion of per-step, and milestone based reward calculation.
- **Training**: Faster and more efficient training, more focus on novel exploration(change in entropy coefficient). Changes in movement of the tactile sensor to encourage more exploration and get more coverage.
- **Testing**: Introduced logging of IoU (coverage) and reward accumulation, thereby making the testing more robust and milestone aware.

V. RESULTS

A. Training Performance



(a) Mean episode reward over training. (b) Mean episode length over training.

Fig. 3: Training performance: The policy improves both reward and episode duration, indicating effective exploration and constraint satisfaction.

When testing on unseen YCB objects, the algorithm achieves strong performance, with IoU (coverage) and reward accumulation showing an upward trend. IoU exceeds 93% on several unknown objects, including Strawberry, Tennis Ball, Tomato Soup Can, Meat Can, Pear, Metal Clamp, and MasterChef Can.

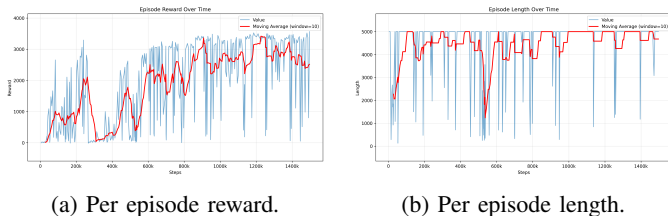


Fig. 4: Fine-grained training performance across episodes.

B. Generalization to Unseen YCB Objects

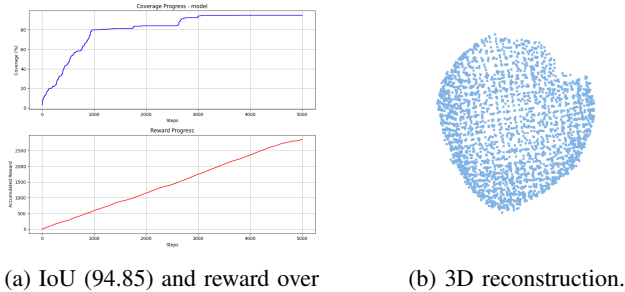


Fig. 5: Performance and 3D reconstruction for the Strawberry object.

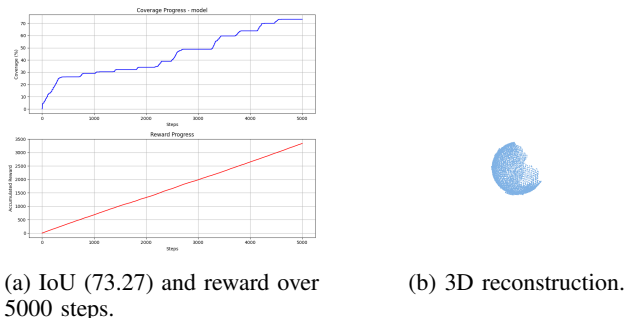


Fig. 6: Performance and 3D reconstruction for the Tennis Ball object.

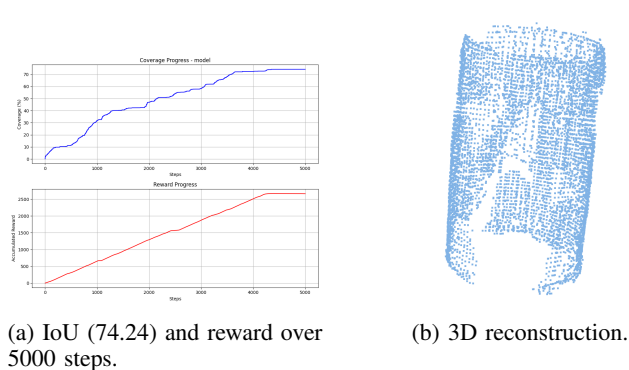


Fig. 7: Performance and 3D reconstruction for the Tomato Soup Can object.

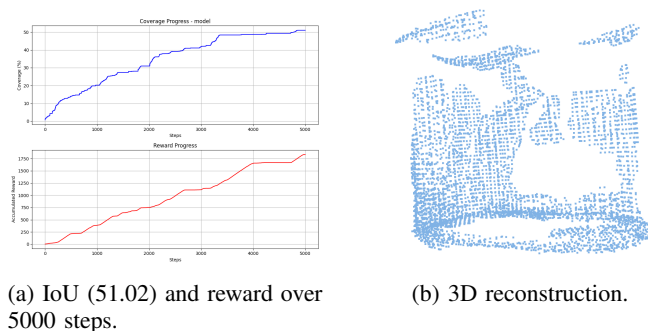


Fig. 8: Performance and 3D reconstruction for the Potted Meat Can object.

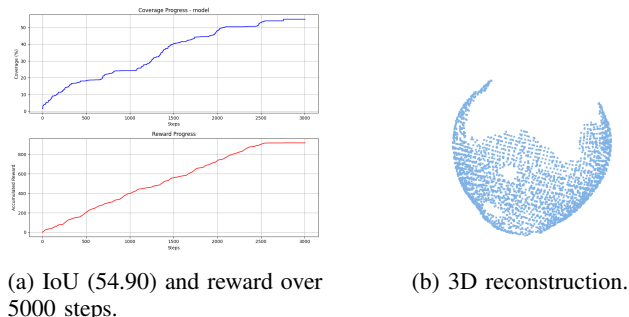


Fig. 9: Performance and 3D reconstruction for the Pear object.

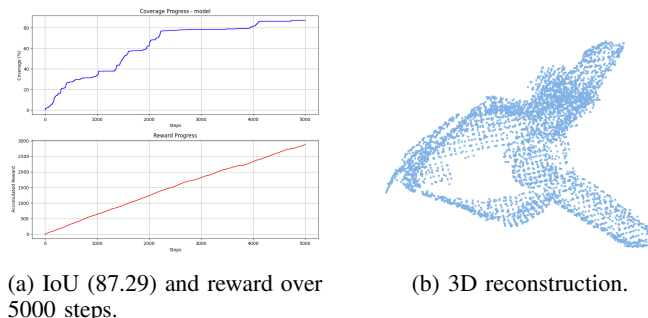


Fig. 10: Performance and 3D reconstruction for the Metal Clamp object.

VI. CONCLUSION

We have successfully implemented a Reinforcement Learning Agent that can actively explore surfaces of unknown objects autonomously. The PPO-based active tactile exploration using a single DIGIT sensor was trained to interact with four distinct primitive objects, and learned an exploration policy that was able to generalize to unseen Ycb objects and generate a corresponding point cloud. The key findings indicate that the RL agent can learn effective strategies for tactile exploration and the agent progressively improved its ability to explore unknown object surfaces. Average IoU over all the objects was over **70%**. There is a scope of improvement in training to improve the reward as in our project, the learning rate was very low, there was a significant plateau of the mean reward was observed. The IoU over a fixed number of 5000 steps can also be improved on certain unseen objects. Early termination is observed on some objects as the agent frequently goes out of bounds.

REFERENCES

- [1] A.-H. Shahidzadeh *et al.*, “AcTExplore: Active Tactile Exploration on Unknown Objects,” *Proc. ICRA*, Jun. 2024.
- [2] S. Wang *et al.*, “TACTO: A fast, flexible and open-source simulator for high-resolution vision-based tactile sensors,” *CoRR*, vol. abs/2012.08456, 2020.
- [3] M. Bauza, A. Bronars, and A. Rodríguez, “Tac2Pose: Tactile Object Pose Estimation from the First Touch,” 2022. [Online]. Available: <https://arxiv.org/abs/2204.01607>
- [4] N. Botteghi, R. Schulte, B. Sirmacek, M. Poel, and C. Brune, “Curiosity-Driven Reinforcement Learning Agent for Mapping Unknown Indoor Environments,” *ISPRS Annals of the Photogrammetry, Remote Sensing and Spatial Information Sciences*, vol. 1, pp. 129–136, 2021.
- [5] G. M. Caddeo, N. A. Piga, F. Bottarel, and L. Natale, “Collision-aware in-hand 6d object pose estimation using multiple vision-based tactile sensors,” in *2023 IEEE International Conference on Robotics and Automation (ICRA)*, pp. 719–725, 2023.
- [6] D. S. Chaplot, D. Gandhi, S. Gupta, A. Gupta, and R. Salakhutdinov, “Learning to Explore using Active Neural SLAM,” *arXiv:2004.05155 [cs]*, 2020.
- [7] F. Chen, S. Bai, T. Shan, and B. Englot, “Self-learning exploration and mapping for mobile robots via deep reinforcement learning,” in *AIAA SciTech 2019 Forum*, p. 0396, 2019.
- [8] F. Chen, J. D. Martin, Y. Huang, J. Wang, and B. Englot, “Autonomous exploration under uncertainty via deep reinforcement learning on graphs,” in *2020 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, pp. 6140–6147, IEEE, 2020.
- [9] K. Ganguly, B. Sadrfaridpour, K. B. Kidambi, C. Fermüller, and Y. Aloimonos, “Grasping in the dark: Compliant grasping using shadow dexterous hand and biotac tactile sensor.”
- [10] D. Gordon, A. Kembhavi, M. Rastegari, J. Redmon, D. Fox, and A. Farhadi, “IQA: Visual question answering in interactive environments,” *CoRR*, abs/1712.03316, 2017.
- [11] F. R. Hogan, M. Jenkin, S. Rezaei-Shoshtari, Y. A. Girdhar, D. Meger, and G. Dudek, “Seeing through your skin: Recognizing objects with a novel visuotactile sensor,” *arXiv preprint arXiv:2011.09552*, 2020.
- [12] H.-J. Liang, N. J. Sanket, C. Fermüller, and Y. Aloimonos, “Salientdso: Bringing attention to direct sparse odometry,” *IEEE Transactions on Automation Science and Engineering*, vol. 16, no. 4, pp. 1619–1626, 2019.
- [13] C. Lin, Z. Lin, S. Wang, and H. Xu, “DTact: A Vision-Based Tactile Sensor that Measures High-Resolution 3D Geometry Directly from Darkness,” *arXiv preprint arXiv:2203.09417*, 2022.
- [14] J. Lin, R. Calandra, and S. Levine, “Learning to Identify Object Instances by Touch: Tactile Recognition via Multimodal Matching,” in *2019 International Conference on Robotics and Automation (ICRA)*, pp. 3644–3650, 2019.
- [15] A. M. Barros, M. Michel, Y. Moline, G. Corre, and F. Carrel, “A comprehensive survey of visual slam algorithms,” *Robotics*, vol. 11, no. 1, 2022.
- [16] U. Martinez-Hernandez, N. F. Lepora, and T. J. Prescott, “Active haptic shape recognition by intrinsic motivation with a robot hand,” in *2015 IEEE World Haptics Conference (WHC)*, pp. 299–304, 2015.
- [17] P. Mirowski, R. Pascanu, F. Viola, H. Soyer, A. J. Ballard, A. Banino, M. Denil, R. Goroshin, L. Sifre, K. Kavukcuoglu, D. Kumaran, and R. Hadsell, “Learning to navigate in complex environments,” *CoRR*, abs/1611.03673, 2016.
- [18] S. Ottenhaus, L. Kaul, N. Vahrenkamp, and T. Asfour, “Active tactile exploration based on cost-aware information gain maximization,” *International Journal of Humanoid Robotics*, vol. 15, p. 1850015, 2018.
- [19] K. Park, H. Yuk, M. Yang, J. Cho, H. Lee, and J. Kim, “A biomimetic elastomeric robot skin using electrical impedance and acoustic tomography for tactile sensing,” *Science Robotics*, vol. 7, no. 67, p. eabm7187, June 2022.
- [20] J. A. Placed, J. Strader, H. Carrillo, N. Atanasov, V. Indelman, L. Carlone, and J. A. Castellanos, “A Survey on Active Simultaneous Localization and Mapping: State of the Art and New Frontiers,” *IEEE Transactions on Robotics*, pp. 1–20, 2023.
- [21] D. Ramani, “A short survey on memory based reinforcement learning,” *arXiv preprint arXiv:1904.06736*, 2019.
- [22] J. Zhao, M. Bauzá, and E. H. Adelson, “Fingerslam: Closed-loop unknown object localization and reconstruction from visuo-tactile feedback,” 2023.